

MUHAMAD SOLEHUDDIN

MANUFACTURING AUTOMATION ENGINEER

Penang, Malaysia | +60 19 228 7350 | saladiniart@gmail.com

linkedin.com/in/solehuddin-muhamad-b67068132 | github.com/SaladinIART | salbotics.uk

PROFESSIONAL PROFILE

Manufacturing automation engineer and Industrial Automation and Robotics engineering technologist with experience spanning production-equipment maintenance, factory digitization, industrial communications, engineering software, and practical troubleshooting. Connects physical equipment to reliable information systems using Modbus, RS-485, MQTT, Python, SQL databases, dashboards, and documented operating workflows. Builds evidence-led demonstrations in MES, machine vision, and robotics while clearly separating deployed work, prototypes, engineering concepts, and work in progress.

CORE ENGINEERING CAPABILITIES

- **Manufacturing automation:** process observation, bottleneck framing, equipment telemetry, PLC and HMI foundations, interlocks, sequencing, and automation trade studies
- **Industrial integration:** Modbus RTU/TCP, RS-485, MQTT, edge Linux devices, Node-RED, Grafana, HMI/SCADA concepts, and OT-to-IT data paths
- **Software and data:** Python, FastAPI, REST APIs, SQL, PostgreSQL, TimescaleDB, Docker Compose, Git, automated tests, dashboards, and technical runbooks
- **Equipment and fabrication:** structured fault isolation, preventive maintenance, electromechanical assembly foundations, SolidWorks, Mastercam, CNC programming, basic industrial wiring, milling, and lathe
- **Emerging engineering areas:** ROS 2 interfaces, computer vision, discrete-event process simulation, AMR/ASRS concepts, MES/ERP workflows, and traceability

PROFESSIONAL EXPERIENCE

Freelance Systems Engineer

Salbotics Solutions | Penang, Malaysia | Oct 2025-Aug 2026

- Built and documented self-directed industrial learning systems covering machine telemetry, APIs, databases, dashboards, PLC/HMI concepts, manufacturing information flow, and local deployment.
- Developed the NEXUS IIoT validated demonstration using Python, FastAPI, MQTT, TimescaleDB, Svelte, Grafana, Docker Compose, automated checks, and documented operating scenarios.
- Built a sanitized MES/ERP manufacturing-flow demonstrator covering synthetic orders, material state, work execution, traceability, quality events, and reporting.
- Produced architecture notes, setup instructions, test evidence, limitations, and troubleshooting runbooks so each project can be reviewed and reproduced with its maturity stated explicitly.

System Engineer, Digitization Department

Alumac Industries Sdn. Bhd. | Sungai Buloh, Malaysia | Jan 2024-Sep 2025

- Developed and supported a real-time factory energy-visibility system across multiple production assets using industrial meters, Modbus RTU over RS-485, Python edge services, SQL storage, and Node-RED dashboards.
- Diagnosed faults across physical signals, serial communications, edge services, databases, and operational dashboards in a live manufacturing environment.
- Integrated a PLC-controlled legacy production asset into an IIoT monitoring path to improve operating visibility and maintenance troubleshooting.
- Translated production, maintenance, management, and IT needs into monitoring logic, decision-facing views, documentation, and handover steps.
- Supported manufacturing reporting and Microsoft 365 workflows using Power BI, SharePoint, Teams, Power Automate, and Power Apps.

Maintenance Technician

Diversatech (M) Sdn. Bhd. | Klang, Malaysia | May 2021-Dec 2023

- Performed diagnostics, preventive maintenance, machine repair, and structured troubleshooting on production equipment.
- Supported engineering teams with fault isolation and recovery while recording machine condition, observed behaviour, and corrective actions for escalation.

CAD and CAM Technician

AMS Seihin Sdn. Bhd. | Kajang, Malaysia | Jun 2016-Sep 2018

- Programmed and troubleshoot CNC equipment, including Brother and HAAS machining centres, using Mastercam and shop-floor verification.
- Used SolidWorks for design and assembly work and supported industrial wiring, milling, lathe, machine repair, and mechanical troubleshooting.

SELECTED ENGINEERING PORTFOLIO

NEXUS Industrial IIoT - Validated Demonstration

- Connected simulated manufacturing signals through Modbus, MQTT, TimescaleDB, FastAPI, Grafana, and a browser decision view; documented reproducible fault and alert scenarios.

MES / ERP Manufacturing Flow - Validated Demonstration

- Modelled synthetic order, material, execution, quality, traceability, and reporting flows in a sanitized manufacturing-information system; not presented as a production MES deployment.

Asset Vision - Prototype

- Built a hardware-independent inspection path using Python, FastAPI, OpenCV, local records, a browser dashboard, typed ROS 2 interfaces, supplied sample images, and automated tests.

Tube Preparation Automation - Engineering Concept

- Decomposed a generic repetitive preparation process into verification, measurement, identification, guarded separation, inspection, output, and kitting modules for an automation trade study.
- Uses invented or generalized inputs only; unvalidated, unaffiliated with any employer, and makes no production, savings, quality, or throughput claim.

EDUCATION AND PROFESSIONAL CREDENTIAL

- **Bachelor of Engineering Technology in Industrial Automation and Robotics Technology** - Universiti Kuala Lumpur Malaysia France Institute (UniKL MFI) | 2017-2023
- **Diploma in Mechatronics** - German-Malaysian Institute
- **Graduate Technologist (GT), Manufacturing and Industrial Technology** - Malaysia Board of Technologists (MBOT) | 2026